ELEC EXAM 1 – MJ1025

INSTRUCTION: **Do not write** anything on this questionnaire. Select the correct answer for each of the following questions. Mark **only one answer** for each item by shading the box corresponding to the letter of your choice on the answer sheet provided.

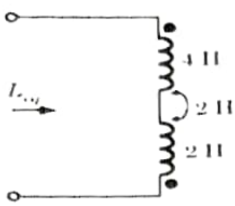
1. If the flow of electric current is parallel to the magnetic field, the force will be ______
A. Zero
B. Infinity
C. Maximum
D. Half the original value

2. Which, among the following qualities, is not affected by the magnetic field?
A. Moving charge
B. Change in magnetic flux
C. Current flowing in a conductor
D. Stationary charge

3. When does a dielectric become a conductor?
A. At avalanche breakdown
B. At high temperature
C. At dielectric breakdown
D. In the presence of magnetic field

4. Find the capacitance of a layer of Al_2O_3 that is 0.5 μm thick and 2000 mm² of square area, εr=8.
A. 1000 μF
B. 0.283 μF
C. 16 μF
D. 2.83 μF

5. For this circuit, Leq = ?



A. 2 H
B. 4 H
C. 6 H
D. 8 H

6. A coil of 360 turns is linked by a flux of 200 μWb. If the flux is reversed in 0.01 second, then find the EMF induced in the coil.
A. 7.2 V
B. 0.72 V
C. 14.4 V
D. 144 V

7. Which one of the following laws will not contribute to the Maxwell's equations?
A. Gauss law
B. Faraday law
C. Ampere law
D. Curie Weiss law

8. Find the curl of E when B is given as 15t.
A. 15
B. -15
C. 7.5
D. -7.5

9. If you have 4 1-ohm resistors and it is desired to have an effective resistance of 1 ohm, how should they be connected?
A. Connect a pair in parallel and then in series with the other two resistors
B. Connect all of them in parallel
C. Connect a pair of them in parallel and then in series with another pair connected in parallel
D. Connect them in series

10. Which has the correct dimension as the electrical potential?
A. A·Ω2
B. N·m/C2
C. kg·m2/(A·s3)
D. J/°C

11. Permeance is the inverse equivalent of which electrical term?
A. Voltage
B. Current
C. Resistance
D. Conductance

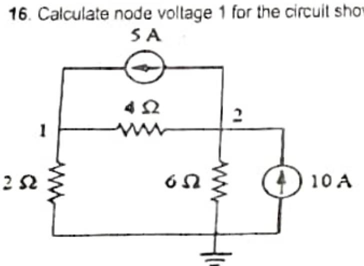
12. What theorem replaces a complex network with an equivalent circuit containing a source voltage and a series resistance?
A. Multinetwork
B. Norton
C. Thevenin
D. Superposition

13. In a series–parallel circuit, individual component power dissipations are calculated using:
A. individual component parameters
B. a percent of the voltage division ratio squared
C. total current squared multiplied by the resistor values
D. a percent of the total power depending on resistor ratios

14. The current flowing through an unloaded voltage divider is called the:
A. resistor current
B. load current
C. bleeder current
D. voltage current

15. When a load is connected to a voltage divider, the total resistance of the circuit will:
A. decrease
B. double
C. increase
D. remain the same

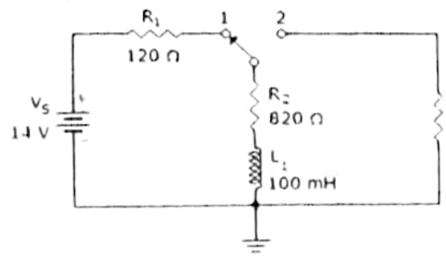
16. Calculate node voltage 1 for the circuit shown:



A. 13.33 V
B. 20 V
C. 23.33 V
D. 33.33 V

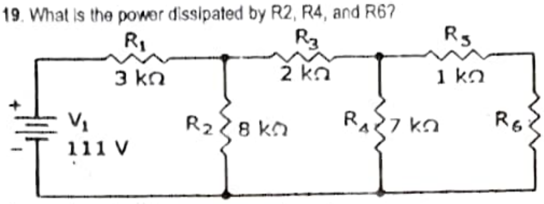
17. Three batteries, each with different internal resistances, has an EMF of 12 V, 9 V and 3 V respectively. If these batteries are connected in parallel, determine the equivalent output voltage assuming 1.2 Ω, 0.3 Ω and 0.2 Ω internal resistances respectively.
A. 3 V
B. 9 V
C. 12 V
D. 6 V

18. In the given circuit, what will the voltage be across R3, 25 μs after the switch is moved to position 2?



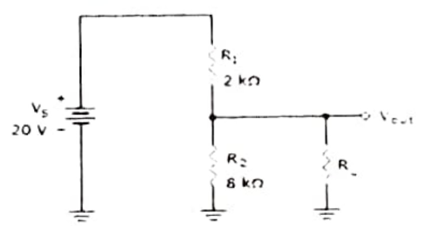
A. 2.88 V
B. 8.34 V
C. 5.9 V
D. 14 V

19. What is the power dissipated by R2, R4, and R6?



A. P2 = 407 mW, P4 = 183 mW, P6 = 156 mW
B. P2 = 397 mW, P4 = 173 mW, P6 = 146 mW
C. P2 = 417 mW, P4 = 193 mW, P6 = 166 mW
D. P2 = 387 mW, P4 = 163 mW, P6 = 136 mW

20. If the load in the given circuit is 80 kΩ, what is the bleeder current?



A. 196 μA
B. 1.96 mA
C. 2 mA
D. 2.16 mA

21. What is the true power of a 24 Vac parallel RL circuit when R = 45 Ω and XL = 1100 Ω?
A. 313.45 W
B. 12.8 W
C. 44.96 W
D. 22.3 W

22. What is the number of turns required in the secondary winding for a transformer when 120 volts is applied to a 2400-turn primary to produce 7.5 Vac at the secondary?
A. 75 turn
B. 900 turn
C. 1920 turn
D. 150 turn

23. What is the inductive reactance if the Q of a coil is 60, and the winding resistance is 5 Ω?
A. 0.083 Ω
B. 12 Ω.
C. 300 Ω.
D. 30 Ω.

24. In a 20 Vac series RC circuit, if 20 V is read across the resistor and 40 V is measured across the capacitor, the applied voltage is
A. 45 Vac
B. 50 Vac
C. 60 Vac
D. 65 Vac

25. How much current will flow in a 100 Hz series RLC circuit if VS = 20 V, RT = 66 Ω, and XT=47Ω?
A. 247 mA
B. 303 mA
C. 1.05 A
D. 107 mA

26. A series resonance network consisting of a resistor of 30Ω, a capacitor of 2uF and an inductor of 20mH is connected across a sinusoidal supply voltage which has a constant output of 9 volts at all frequencies. Calculate the resonant frequency
A. 769Hz
B. 796Hz
C. 967Hz
D. 976Hz

27. A parallel resonance network consisting of a resistor of 60Ω, a capacitor of 120uF and an inductor of 200mH is connected across a sinusoidal supply voltage which has a constant output of 100 volts at all frequencies. Calculate the quality factor
A. 7.41
B. 4.17
C. 1.47
D. 7.14

28. A series RLC circuit containing a resistance of 12Ω, an inductance of 0.15H and a capacitor of 100uF are connected in series across a 100V, 50Hz supply. Calculate the total circuit impedance
A. 47.13 Ω
B. 31.83 Ω
C. 19.4 Ω
D. 5.14 Ω

29. A 1kΩ resistor, a 142mH coil and a 160uF capacitor are all connected in parallel across a 240V, 60Hz supply. Calculate the current drawn from the supply.
A. 10A
B. 20A
C. 30A
D. 40A

30. Y parameter called short circuit output admittance
A. y11 B. y12 C. y21 D. y22

31. A wound coil that has an inductance of 180mH and a resistance of 35Ω is connected to a 100V 50Hz supply. Calculate the apparent power consumed.
A. 100VA B. 150VA C. 200VA D. 250VA

32. A sinusoidal waveform is defined as: Vm = 169.8 sin(377t) volts. Calculate the RMS voltage of the waveform
A. 169.8Vrms B. 120Vrms C. 130.8Vrms D. 129.6Vrms

33. You are testing a transistor. One lead is on the base and the other is on the emitter. The DMM showed either high or low resistance for both polarities. From this observation, you can therefore conclude that the transistor is ______
A. npn B. pnp C. faulty D. working

34. You were testing a diode in a circuit using a DMM. You put the red probe on the lead part with a marking, and the black probe on the other lead. The DMM showed a reading of -0.6 V. What can you say about the biasing of the diode, and about the placing of the DMM probes?
A. Reverse biased, reversed polarity
B. Reverse biased, proper polarity
C. Forward biased, reversed polarity
D. Forward biased, reversed polarity

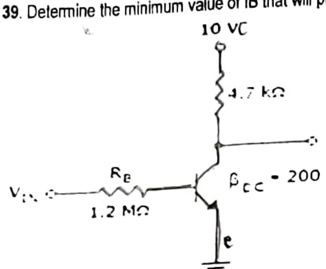
35. Which of the following BJT terminals could be used as an input port?
A. Base and Collector B. Base and Emitter
C. Collector and Emitter D. All of the above

36. The ______ configuration is frequently used for impedance matching.
A. fixed bias B. voltage-divider bias
C. emitter-follower D. collector feedback

37. Calculate the ripple of a capacitor filter for a peak rectified voltage of 30 V, capacitor C= 100μF, and a load current of 50 mA.
A. 3.7% B. 4.3% C. 5.1% D. 6.9%

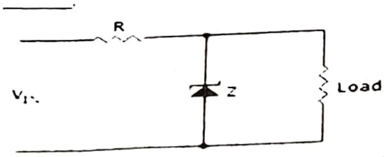
38. A JFET
A. is a current-controlled device B. has a low input resistance
C. is a voltage-controlled device D. is always forward-biased

39. Determine the minimum value of IB that will produce saturation.



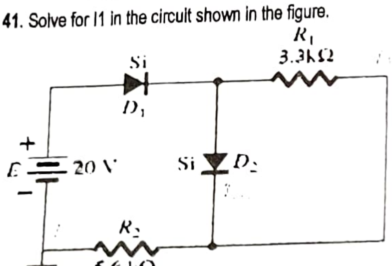
A. 0.25 mA B. 5.325 A C. 1.065 μA D. 10.425 μA

40. Refer to this figure. If the load current increases, IR will ______ and IZ will ______.



A. remain the same, increase B. decrease, remain the same
C. increase, remain the same D. remain the same, decrease

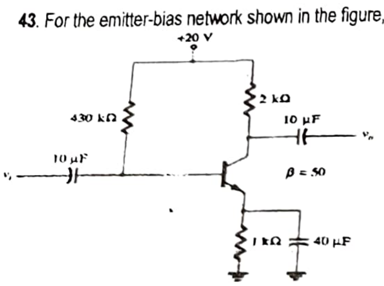
41. Solve for I1 in the circuit shown in the figure.



A. 0.311 mA B. 0.212 mA C. 3.11 mA D. 2.12 mA

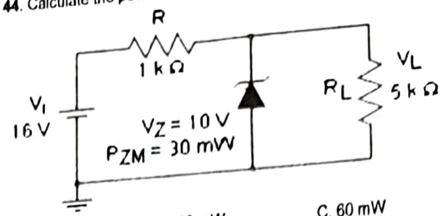
42. Solve for the dynamic ac resistance of a diode whose current is 25 mA.
A. 1.04 Ω B. 1.15 Ω C. 0.53 Ω D. 2.12 Ω

43. For the emitter-bias network shown in the figure, determine IB.



A. 20 μA B. 15 μA C. 35 μA D. 40 μA

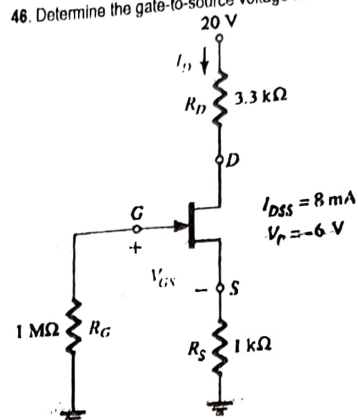
44. Calculate the power dissipated across the Zener diode in the circuit shown below:



A. 20 mW B. 40 mW C. 60 mW D. 80 mW

45. Which of the following conditions must be met to allow the use of the approximate approach in a voltage-divider bias configuration?
A. βre > 10R2 B. βRE > 10R2 C. βRE < 10R2 D. βre < 10R2

46. Determine the gate-to-source voltage for the circuit shown in the figure.

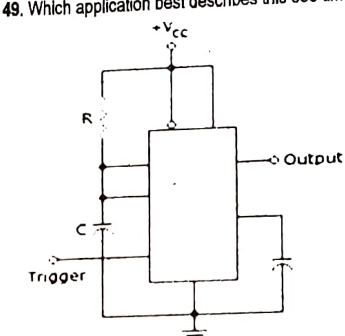


A. -2.6 V B. 3.2 V C. -2.8 V D. 3.4 V

47. A voltage-follower amplifier comes to you for service. You find the voltage gain to be 5.5 and the input impedance 22kΩ. The probable fault in this amplifier, if any, is
A. the gain is too low for this type of amplifie
B. the input impedance is too high for this amplifie
C. nothing is wron
D. none of these

48. What do you call an oscillator that uses a transformer in parallel with a capacitor?
A. Armstrong oscillator B. Clapp oscillator
C. Colpitts oscillator D. Hartley oscillator

49. Which application best describes this 555 timer circuit?

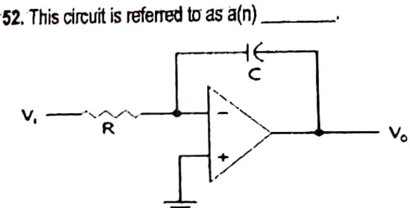


A. Monostable multivibrator B. Astable multivibrator
C. Bistable multivibrator D. Free-running multivibrator

50. Calculate the frequency of oscillation of the phase-shift oscillator with gain of 29, R = 1 kiloohms and C = 1 microfarad.
A. 26 Hz B. 65 Hz C. 135 Hz D. 238 Hz

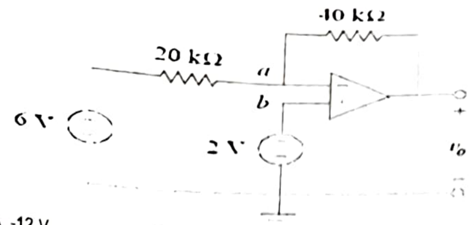
51. Three-terminal fixed positive voltage regulators commonly used in industry.
A. 78XX series B. 79XX series C. 723 IC regulator D. 317 regulator

52. This circuit is referred to as a(n) ______.



A. inverting amplifier B. noninverting amplifier
C. differentiator D. integrator

53. Determine vo in the ideal op-amp circuit shown in the figure.,



A. -12 V B. 4 V C. 6 V D. -4 V

54. The attenuation of the three-section RC feedback phase-shift oscillator is
A. 1/9. B. 1/30. C. 1/3. D. 1/29.

55. Determine the output impedance with feedback for a voltage-series feedback having A = −100, R1 = 15 kΩ, Ro = 20 kΩ, and a feedback of β = −0.25.
A. 0.2 kΩ B. 392.16Ω C. 1.82 kΩ D. 769.23Ω

56. Calculate the input bias currents at the non-inverting and inverting terminal respectively of an op-amp having specified values of I_IO=5nA and I_IB=30nA
A. 32.5nA;32.5nA B. 27.5nA;27.5nA C. 27.5nA;32.5nA D. 32.5nA;27.5nA

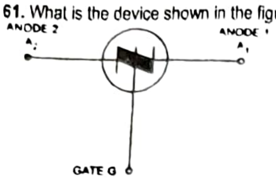
57. Determine the output voltage of an op-amp for input voltages of V_I1=150μV and V_I2=140μV. The amplifier has a differential gain of A_d=4000 and the value of the CMRR is 100.
A. 32.6mV B. 45.8mV C. 51.9mV D. 66.7mV

58. What is the response of a Twin-T oscillator?
A. bandstop B. high – pass C. low – pass D. bandpass

59. What is the input bias current of an LM741C operational amplifier?
A. 90 nA B. 25 nA C. 80 nA D. 30 nA

60. The maximum rate of change of the output voltage in response to a step input voltage is the ________ of an op-amp.
A. time constant B. maximum frequency
C. slew rate D. static discharge

61. What is the device shown in the figure?



A. Silicon Unilateral Switch B. Silicon Controlled Switch
C. Silicon Bilateral Switch D. Silicon Controlled Rectifier

62. How do you change the angle of conduction in an SCR circuit?
A. Tilt the SC B. Vary the cathode voltage
C. Vary the anode voltage D. Vary the gate voltage

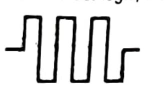
63. Zero-voltage switching is commonly used in ____.
A. RFI generation B. Balanced bridge circuits
C. SCR and triac power-control circuits D. Determining thermocouple voltage

64. What is also known as a bidirectional thyristor and is equivalent to 2 Shockley diode connected to allow current in either direction?
A. Triac B. SCR C. JFET D. Diac

65. RTDs are typically connected with other fixed resistors
A. in a pi configuration B. in a bridge configuration
C. and variable resistors D. and capacitors in a filter-type circuit

66. What is used in conjunction with security systems to generate video signals, control and transmit/display video signals, and real time storing of video signals/alarm events.
A. CCTV B. CATV C. DTV D. SETV

67. In ladder logic, the given symbol is used to indicate

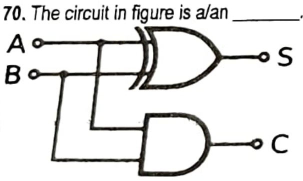


A. fuse B. heating element C. relay coil D. flow switch

68. Determine RB1 for a silicon PUT if it is determined that h = 0.84, VP = 11.2 V, and RB2 = 5 kΩ.
A. 12.65 kΩ B. 16.25 kΩ C. 20.00 kΩ D. 26.25 kΩ

69. Which of the following is not an output of a tristate logic?
A. High B. High Z C. Low D. Low Z

70. The circuit in figure is a/an ________.



A. Half-adder B. Full-adder C. Full Subtractor D. Half Subtractor

71. What logic gate can be realized from the truth table shown in Figure?

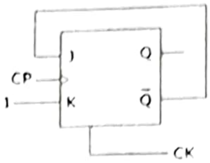
A	B	C	Y
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

A. Two input OR gate B. Three input OR gate
C. Two input AND gate D. Three input AND gate

72. When does the output of a NOR gate = 0?
A. Whenever a 0 is present at an input B. Only when all inputs = 0
C. Whenever a 1 is present at an input D. Only when all inputs = 1

73. Express the decimal number −37 as an 8-bit number in sign-magnitude.
A. 10100101 B. 00100101 C. 11011000 D. 11010001

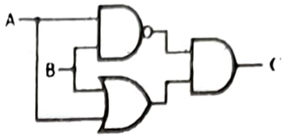
74. In a J-K FF we have J = Q and K = 1. Assuming the FF was initially cleared and then clocked for 6 pulses, the sequence at the Q output will be



A. 010000 B. 011001 C. 010010 D. 010101

75. Convert binary number 11000110 to gray code ________
A. 10100111 B. 10000101 C. 10111011 D. 10100101

76. What is the equivalent gate of the circuit below?



A. NAND B. NOR C. XOR D. XNOR

77. Encode the decimal number 45 in BCD code.
A. 0100 0110 B. 0101 0101 C. 0100 0100 D. 0100 0101

78. The systematic reduction of logic circuits is accomplished by:
A. symbolic reduction B. TTL logic
C. using Boolean algebra D. using a truth table

79. Made up from basic logic NAND, NOR or NOT gates that are connected together to produce more complicated switching circuits.
A. Combinational Logic Circuits B. Sequential Logic Circuits
C. Algorithmic State Machine D. Asynchronous Sequential Logic

80. If the program is placed in ROM or EPROM, it referred to as ____.
A. software B. firmware C. hardware D. software and firmware

81. A TTL NAND gate with IL(max) of -1.6mA per input drives eight TTL inputs. How much current does the drive output sink?
A. -12.8 mA B. -8 mA C. -1.6 mA D. -200 uA

82. Which method bypasses the CPU for certain types of data transfer?
A. Software interrupts B. Interrupt-driven I/O
C. Polled I/O D. Direct memory access (DM

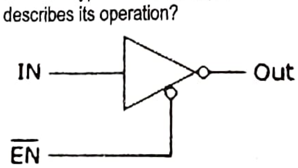
83. Which of the following is not an enhancement to the Pentium that was unavailable in the 8086/8088?
A. "Pipelined" architecture B. Expansion of cache memory
C. Inclusion of an Internal math coprocessor D. Data/address line multiplexing

84. Which bus is bidirectional?
A. Address bus B. Control bus. C. Data bus D. None of the above

85. Which is not part of the execution unit (EU)?
A. Arithmetic logic unit (AL B. Clock C. General registers D. Flags

86. Which of the following is not a jump instruction?
A. JB (jump back) B. JA (jump above)
C. JO (jump if overflow) D. JMP (unconditional jump)

87. What type of circuit is represented in the given figure, and which statement best describes its operation?



A. It is a tristate inverter
B. It is a programmable inverter
C. It is an active LOW buffer, which can be turned on and off by the ENABLE input
D. None of the above

88. The IEEE/ANSI notation of an internal underlined diamond denotes:
A. totem-pole output B. open-collector output
C. quadrature amplifier D. tristate buffer

89. Polled I/O works best when ______.
A. there are no priority considerations
B. priority considerations are frequent
C. the polling rate exceeds 1000s
D. the polling rate is below 1000s

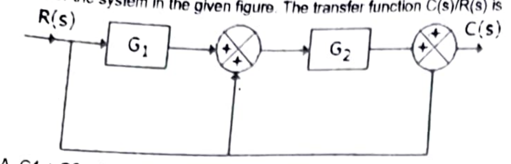
90. Proper handling of a CMOS device is necessary because of its
A. fragile construction
B. high-noise immunity
C. susceptibility to electrostatic discharge
D. low power dissipation

91. A certain gate draws 1.8 μA when its output is HIGH and 3.3 μA when its output is LOW. VCC is 5 V and the gate is operated on a 50% duty cycle. What is the average power dissipation (PD)?
A. 12.75 μW B. 2.55 μW C. 1.27 μW D. 5 μW

92. Which equation is correct?
A. VNL = VIL(max) B. VNH = VOH(max) C. VNL = VOH(min) D. VNH = VOH(min)

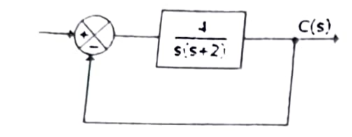
93. In an integral controller
A. the output is proportional to input
B. the rate of change of output is proportional to input
C. the output is proportional to rate of change of input
D. none of the above

94. For the system in the given figure. The transfer function C(s)/R(s) is



A. G1 + G2 + 1 B. G1 G2 + 1 C. G1 G2 + G2 + 1 D. G1 G2 + G1 + 1

95. For the system of the given figure the closed loop poles are located at



A. s = 0 and s = -2 B. s = 0, s = -1 ± j√3 C. s = -1 ± j√3 D. s = -2 and s = -1 ± j√3

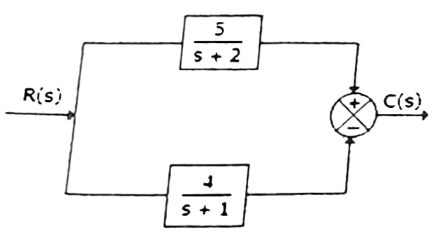
96. For which systems are the signal flow graphs applicable?
A. Causal B. Invertible C. Linear time invariant system D. Dynamic

97. What are the two basic element of signal flow graph?
A. Nodes and modes
B. Nodes and branches
C. loops and branches
D. loops and nodes

98. Finding a network that represents a given transfer function.
A. Transient analysis
B. Stability analysis
C. Steady state analysis
D. Network synthesis

99. Which type of node comprises incoming as well as outgoing branches?
A. Source node B. Sink node C. Chain node D. Main node

100. The system of the given figure



A. has two poles and one zero in left half plane
B. has two poles and one zero in right half plane
C. has two poles in left half plane and zero in right half plane
D. none of the above

101. High-voltage power transmission lines experience greater losses at higher frequencies due to:
A. Hysteresis effect
B. Skin effect
C. Eddy currents in air
D. Displacement current

102. A coaxial cable carrying RF signals is designed primarily to:
A. Prevent external interference and confine fields
B. Increase capacitance of the circuit
C. Radiate energy efficiently
D. Reduce dielectric breakdown

103. The energy flow of an antenna in free space is best represented by:
A. Electric field intensity
B. Magnetic flux density
C. Poynting vector
D. Permittivity of space

104. The permittivity of a dielectric material in a capacitor directly affects:
A. Induced EMF
B. Stored energy
C. Magnetic field strength
D. Current density in conductors

105. Which Maxwell's equation explains how a transformer induces voltage in its secondary winding?
A. Gauss's Law for electricity
B. Gauss's Law for magnetism
C. Faraday's Law of Induction
D. Ampere's Law with Maxwell's correction

106. In waveguides used for microwave transmission, the dominant TE10 mode has a cutoff frequency determined by:
A. Width of the guide
B. Height of the guide
C. Length of the guide
D. Dielectric loss tangent

107. Circular polarization of an electromagnetic wave is achieved when:
A. Electric and magnetic fields are perpendicular
B. Electric field components are equal in magnitude and 90° out of phase
C. Current and voltage are in phase
D. Magnetic field is aligned with propagation direction

108. The boundary condition that makes Wi-Fi signals weaker inside reinforced concrete buildings is mainly due to:
A. Low permeability of concrete
B. High permittivity of air
C. Reflection and absorption by materials
D. Displacement current in steel rods

109. A car battery rated at 12 V is modeled as an ideal voltage source with internal resistance. When headlights are turned on, the observed voltage at the terminals drops slightly. This drop is due to:
A. Battery's capacitance
B. Internal resistance
C. Short circuit protection
D. Temperature variation

110. A PCB trace carrying DC current heats up noticeably. The heating is directly proportional to:
A. Current B. Current squared times resistance C. Voltage D. Power factor

111. The effective resistance of two unequal resistors in parallel is
A. Always greater than the smaller resistor
B. Always less than the smaller resistor
C. Equal to the average of the two
D. Equal to their product

112. A supercapacitor connected across a DC supply is used in electronics mainly to:
A. Reduce AC harmonics
B. Provide quick bursts of stored energy
C. Replace resistors in biasing circuits
D. Increase battery voltage

113. The time constant of an RL circuit defines:
A. The time it takes for current to drop to zero
B. The rate of change of flux
C. The speed at which current rises after switch closure
D. The resistance-to-inductance ratio

114. In household wiring, a "ground loop" can cause voltage differences between equipment grounds. This is due to:
A. Pure inductance in wires
B. Finite resistance of conductors
C. Capacitance between devices
D. Magnetic shielding failure

115. Thevenin's theorem is useful in DC circuits primarily for:
A. Minimizing node voltages
B. Replacing a network with an equivalent voltage source and resistance
C. Reducing the need for capacitors
D. Determining inductance in loops

116. A dependent current source in a DC model of a transistor is controlled by:
A. Its own resistance
B. A voltage or current elsewhere in the circuit
C. The applied supply voltage only
D. Ambient temperature

117. The stored energy in a capacitor in DC circuits is given by:
A. $W = \frac{1}{2}CV^2$ B. $W = \frac{1}{2}LI^2$ C. $W = I^2R$ D. $W = VI$

118. When applying Superposition theorem in DC circuits:
A. Dependent sources are turned off
B. Independent sources are turned off one at a time
C. Only current sources are kept active
D. It is valid only for reactive elements

119. The efficiency of a DC power supply is limited by:
A. Size of resistors
B. Losses due to internal resistance and heat
C. Use of dependent sources
D. Frequency of switching

120. In nodal analysis, the reference node (ground) is chosen because:
A. It simplifies capacitor charging equations
B. It reduces the number of simultaneous equations by one
C. It eliminates resistors in parallel
D. It ensures equal current distribution

121. In a fluorescent lighting system, a capacitor is connected in parallel to the ballast. Its main purpose is to:
A. Store excess charge
B. Correct the lagging power factor
C. Reduce flicker frequency
D. Increase inductance of the ballast

122. In a pure inductive AC circuit, the average real power over one cycle is:
A. Zero
B. Maximum
C. Negative
D. Equal to reactive power

123. The current in a household appliance lags the voltage. This indicates that the load is:
A. Capacitive B. Resistive C. Inductive D. Nonlinear

124. In audio crossover networks, capacitors are used in series with tweeters because they:
A. Block high frequencies
B. Block low frequencies
C. Amplify midrange signals
D. Reduce total resistance

125. At resonance in a series RLC circuit, the circuit current is limited only by:
A. Inductance
B. Resistance
C. Capacitance
D. Reactance

126. The impedance of a parallel RC circuit at very high frequency approaches:
A. Zero B. Infinity C. Pure resistance D. Pure capacitance

127. The power factor of an AC system is improved by:
A. Adding inductors in series
B. Adding capacitors in parallel

C. Increasing resistance of loads
D. Reducing supply voltage

128. The RMS value of a sinusoidal current is related to its peak value by
A. IRMS = Ipeak / 2
B. IRMS = Ipeak / √2
C. IRMS = Ipeak × √2
D. IRMS = Ipeak²

129. The bandwidth of a resonant RLC circuit is related to quality factor Q by:
A. BW = f0 × Q B. BW = f0 / Q C. BW = Q / f0 D. BW = f0² × Q

130. In Thevenin's theorem applied to AC circuits, the equivalent impedance may include:
A. Resistance only
B. Reactance only
C. Complex impedance with both resistance and reactance
D. No impedance at all

131. In a Norton equivalent for AC, the current source value is determined by:
A. Open-circuit test
B. Short-circuit current
C. Voltage divider
D. Frequency response

132. The ABCD parameters of a two-port network are especially useful in analyzing:
A. Power factor correction
B. Transmission lines
C. Parallel resonance circuits
D. Transformer losses

133. A silicon diode in forward bias conducts significantly only after about:
A. 0.1 V B. 0.3 V C. 0.7 V D. 1.2 V

134. A Zener diode used in reverse bias is ideal for:
A. Amplification
B. Voltage regulation
C. Power rectification
D. Frequency doubling

135. A Schottky diode is preferred in digital logic circuits because it:
A. Has high breakdown voltage
B. Switches very fast with low forward drop
C. Works only in reverse bias
D. Has large junction capacitance

136. In a full-wave rectifier, the output frequency is:
A. Same as input AC
B. Twice the input AC frequency
C. Half the input AC frequency
D. Zero

137. A tunnel diode is unique because it:
A. Operates only in forward bias
B. Exhibits negative resistance due to quantum tunneling
C. Has high forward resistance
D. Has no depletion region

138. The main reason LEDs emit visible light while silicon diodes do not is:
A. LEDs are thinner
B. LED materials have direct bandgap
C. LEDs need less current
D. LEDs have lower threshold

139. A photodiode is usually operated in reverse bias because:
A. Forward bias increases leakage
B. Reverse bias widens depletion region and improves sensitivity
C. Reverse bias reduces capacitance to zero
D. Forward bias blocks light response

140. In a BJT, the active region is characterized by:
A. Both junctions reverse biased
B. Both junctions forward biased
C. Emitter-base forward biased, collector-base reverse biased
D. Emitter-base reverse biased, collector-base forward biased

141. The cutoff region of a transistor corresponds to:
A. High collector current
B. ~~Both junctions reverse biased~~
C. Saturation of collector current
D. Negative resistance

142. A MOSFET differs from a JFET because:
A. It has a forward-biased gate
B. Its gate is insulated by oxide, giving very high input impedance
C. It requires large gate current
D. ~~It cannot be used in switching~~

143. The threshold voltage of a MOSFET is defined as the minimum:
A. Source-to-drain voltage to start conduction
B. Gate-to-source voltage to form a conductive channel
C. Drain-to-source current measurable
D. Base current for conduction

144. A varactor diode is commonly used in:
A. Power regulation
B. Frequency tuning in RF circuits
C. Full-wave rectification
D. LED displays

145. The maximum efficiency of a half-wave rectifier is about:
A. 25% B. 40.6% C. 50% D. 81.2%

146. A voltage doubler circuit increases:
A. Output current
B. Output power
C. Output voltage peak to about twice the input
D. Input frequency

147. In audio amplifiers, bypass capacitors are used across emitter resistors primarily to:
A. Reduce bias current
B. Maintain AC gain while keeping DC stability
C. Increase input resistance
D. Prevent saturation

148. A common-collector amplifier is widely used in buffer stages because it:
A. ~~Provides very high voltage gain~~
B. Provides low output impedance with unity gain
C. Provides negative resistance at the output
D. Blocks DC and passes AC only

149. The gain-bandwidth product of an op-amp is:
A. Constant regardless of gain setting
B. Dependent on supply voltage
C. Zero at high frequency
D. Equal to maximum open-loop gain

150. In high-speed op-amps, slew rate limits:
A. Maximum DC gain
B. Maximum rate of change of output voltage
C. Minimum input bias current
D. Thermal stability

151. A Darlington pair is used in amplifier design to:
A. Reduce base current and increase current gain
B. Increase cutoff frequency
C. Act as a negative feedback loop
D. Eliminate saturation voltage

152. In a tuned amplifier, using a high-Q tank circuit:
A. Increases bandwidth
B. Narrows bandwidth and improves selectivity
C. Reduces signal distortion
D. Increases resistive losses

153. In practical op-amp circuits, input offset voltage causes:
A. Oscillations at resonance
B. Nonzero output even when both inputs are zero
C. Increased gain
D. Reduced slew rate

154. A Schmitt trigger circuit is mainly used to:
A. Amplify low-frequency signals
B. Remove noise by introducing hysteresis
C. Act as a passive filter
D. Reduce DC offset

155. A phase-shift oscillator requires three RC sections because:
A. Each section provides 120° phase shift
B. Each section provides 60° phase shift
C. Each section reduces distortion
D. Each section increases loop gain

156. Wien bridge oscillators typically require an automatic gain control (like a lamp or nonlinear resistor) to:
A. Set frequency
B. Prevent distortion by stabilizing amplitude
C. Reduce power factor
D. Generate harmonics

157. Negative feedback in op-amp design can unintentionally cause oscillation when:
A. Phase shift approaches 180° with gain ≥ 1
B. Gain is reduced below unity
C. Bandwidth is increased
D. DC biasing is stable

158. A notch filter in audio systems is used to:
A. Pass only high frequencies
B. Reject a very narrow unwanted frequency like 60 Hz hum
C. Improve gain-bandwidth product
D. Reduce amplifier noise floor

159. A class AB amplifier is used in audio systems instead of pure class A because it:
A. ~~Eliminates crossover distortion completely~~
B. Balances efficiency with acceptable distortion
C. Provides infinite input impedance
D. Needs no biasing

160. The cutoff frequency of a simple Wien Bridge is:
A. f = RC
B. f = 1 / RC
C. f = 1 / (2πRC)
D. f = 2πRC

161. In elevator motor drives, SCR-based controllers are used instead of mechanical contactors because they:
A. Eliminate need for braking
B. Provide smooth speed control via phase angle adjustment
C. Reduce stray capacitance
D. Operate only with DC supply

162. A TRIAC-based dimmer for LED lamps often requires additional circuitry because:
A. TRIACs cannot switch AC directly
B. LED drivers may misfire due to non-sinusoidal current
C. TRIACs conduct only DC
D. LEDs require high inductance for proper dimming

163. A DIAC in lamp dimmer circuits ensures:
A. Higher conduction angle
B. Symmetrical triggering of the TRIAC in both half-cycles
C. Reduced DC offset
D. Direct current blocking

164. In smoke detection systems, photodiodes are often used in reverse bias to:
A. Increase leakage current
B. Act as a fast light sensor with wide depletion region
C. Produce constant voltage
D. Block IR radiation

165. In PLC-based factory automation, ladder logic is widely used because it:
A. Eliminates wiring requirements
B. Resembles relay control diagrams, making it intuitive for technicians
C. Requires no programming device
D. Provides higher switching speed than microcontrollers

166. An LDR-based automatic streetlight turns on at night because:
A. Its resistance decreases with darkness
B. Its resistance increases with darkness, causing a control circuit to switch the lamp
C. It generates a constant current
D. It stores energy during the day

167. A PIR sensor in smart lighting systems primarily detects:
A. Changes in magnetic flux
B. Variations in infrared radiation from moving warm objects
C. Sound levels above threshold
D. Voltage drop across resistors

168. In modern building management systems (BMS), optocouplers are used in sensor interfaces to:
A. Increase amplifier gain
B. Isolate high-voltage equipment from low-voltage control circuits
C. Improve time constant of RC circuits
D. Reduce relay noise

169. In digital electronics, hazards (glitches) in combinational circuits occur due to:
A. Excessive fan-out
B. Unequal propagation delays of logic gates
C. Using only NAND gates
D. Wrong Boolean simplification

170. A BCD-to-decimal decoder is most likely used in:
A. Arithmetic logic units B. Seven-segment display drivers
C. Memory addressing D. Error correction circuits

171. A parity bit in digital communication is used to:
A. Detect single-bit errors B. Correct multiple errors
C. Increase clock speed D. Improve bandwidth

172. In Boolean algebra, the expression A + AB simplifies to:
A. AB B. A C. B D. A + B

173. Which flip-flop is commonly used to eliminate the race-around problem in JK designs?
A. RS flip-flop B. T flip-flop C. Master–slave JK flip-flop D. D flip-flop

174. The main advantage of using Gray code in position encoders is:
A. It reduces hardware cost
B. Only one bit changes at a time, minimizing errors
C. It allows direct binary arithmetic
D. It doubles counting speed

175. A synchronous counter is preferred over an asynchronous counter because it:
A. Consumes less power
B. Has no propagation delay accumulation
C. Needs no clock input
D. Works only in ripple mode

176. A multiplexer can function as:
A. A universal combinational circuit B. A latch replacement
C. A memory element D. A pulse generator

177. A ring counter with 4 flip-flops has how many unique states?
A. 4 B. 8 C. 10 D. 16

178. In a programmable logic array (PLA), both AND and OR arrays are:
A. Fixed
B. Programmable
C. Nonexistent
D. Used only for sequential circuits

179. An algorithmic state machine (ASM) chart is better than a state diagram because it:
A. Eliminates sequential circuits
B. Integrates flowchart and state representation for clarity
C. Provides Boolean simplification directly
D. Works only with asynchronous systems

180. A hazard-free circuit design can be achieved by:
A. Increasing fan-in of gates
B. Adding redundant logic terms in K-map simplification
C. Using only XOR gates
D. Lowering supply voltage

181. In a microprocessor, the program counter is updated:
A. Before instruction fetch B. During instruction decode
C. After fetching an instruction D. Only during branch instructions

182. A stack overflow in microcontrollers usually occurs when:
A. The program counter exceeds 16 bits
B. Too many subroutine calls or interrupts use limited stack memory
C. Data bus is overloaded with writes
D. Input ports are left floating

183. The accumulator is central in 8-bit microcontrollers because it:
A. Generates clock pulses
B. Stores intermediate results of arithmetic and logic operations
C. Addresses external memory
D. Controls I/O devices directly

184. Which memory is best suited for storing firmware in embedded systems?
A. SRAM B. DRAM C. Flash ROM D. Cache

185. A watchdog timer is included in many microcontrollers to:
A. Save power during idle states
B. Reset the system if the program becomes unresponsive
C. Provide real-time clock functionality
D. Generate non-maskable interrupts

186. The control bus in a microprocessor system carries:
A. Actual data values
B. Timing and command signals like Read/Write
C. Memory addresses
D. Error codes from ALU

187. Microcontrollers are preferred over microprocessors in embedded designs because they:
A. Have faster instruction sets
B. Integrate CPU, memory, and I/O on one chip
C. Require external RAM for all programs
D. Are designed only for desktop applications

188. Immediate addressing mode means that:
A. The instruction specifies the memory address directly
B. The instruction contains the actual operand value
C. The operand is stored in the accumulator
D. The operand address is held in another register

189. The width of the data bus determines:
A. The maximum memory addressable
B. How many bits can be transferred in parallel per operation
C. The number of I/O ports available
D. The size of program counter

190. ARM processors are widely used in smartphones because they:
A. Operate only in 8-bit mode
B. Consume low power with high performance (RISC design)
C. Require external floating-point units
D. Are cheaper because they lack interrupts

191. In assembly programming, the instruction CALL 2000H means:
A. Jump to location 2000H without saving return address
B. Jump to subroutine at 2000H and store return address on stack
C. Load 2000H into accumulator
D. Reset program counter to zero

192. The zero flag in the status register is set when:
A. The ALU result is exactly zero B. An overflow occurs
C. A carry is generated D. An interrupt is pending

193. In cruise control systems of cars, negative feedback is used to:
A. Eliminate damping
B. Stabilize speed against load variations
C. Increase open-loop gain indefinitely
D. Reduce sensitivity to throttle input

194. In a block diagram, the purpose of a summing junction is to:
A. Multiply signals
B. Compare reference input and feedback to generate error
C. Act as a low-pass filter
D. Provide phase lead

195. A Type-0 control system subjected to a ramp input will exhibit:
A. Zero steady-state error
B. Constant finite steady-state error
C. Infinite steady-state error
D. Oscillatory error

196. In root locus analysis, adding a zero to the open-loop transfer function generally:
A. Moves the poles farther left, improving transient response
B. Makes the system unstable automatically
C. Reduces system gain to zero
D. Increases the number of poles

197. In the force–voltage analogy, mechanical spring stiffness corresponds to:
A. Inductance B. Resistance C. Capacitance D. Voltage source

198. A control system with very small gain margin is likely to:
A. Be overdamped
B. Exhibit sustained oscillations or instability
C. Have zero steady-state error
D. Operate as critically damped

199. Bode plots are useful in control system design because they:
A. Show system time constants directly
B. Provide frequency-domain stability margins (gain and phase)
C. Eliminate need for root locus
D. Indicate system bandwidth only

200. Lag compensation is often applied in industrial control systems to:
A. Improve steady-state accuracy at the expense of speed
B. Increase bandwidth and transient response
C. Remove integrator action completely
D. Eliminate feedback entirely